Response to the Major-Revision Examiner Report

Gaussian-Process Reconstruction of the Mapping-QCLE Excess Term: A Moving-Cloud Formulation and Failure Analysis

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Revision overview

The revision presents the work as a controlled negative-result study. Accurate density interpolation is separated from operator fidelity, SEO structure, raw conservation, stochastic stability, and physical accuracy. Because these requirements are not jointly satisfied, the revised thesis does not call the MIDPOINT construction reliable and does not claim a systematic improvement over PBME.

Final mandatory corrections

1 — Time-step inventory reconciled

Status: Implemented and verified in the revised source

Correction: The campaign is stated as 24 paired momentum–seed–step configurations, 48 individual method executions, and four independent seeds.

Final location: Thesis p. 80, Section 5 time-step tests; Table G.4.

Archive evidence: final_reviewer_closure/timestep/timestep_run_by_run.csv

2 — Reference settings synchronized

Status: Implemented and verified in the revised source

Correction: Table 6.12 is generated from the same full-precision CSVs as Table F.1 and lists momentum, final time, all domain and grid/time levels, physical edge statistic, and a defined CFL ratio.

Final location: Thesis p. 102, Table 6.12; Appendix F, Table F.1.

Archive evidence:
final_reviewer_closure/reference_settings_by_method_and_momentum.csv

3 — Temporal orders corrected

Status: Implemented and verified in the revised source

Correction: The text assigns approximately second-order temporal behaviour to TDSE and approximately fourth-order temporal behaviour to grid-QCLE RK4; spatial/grid behaviour is treated separately.

Final location: Thesis p. 100, Section 6 reference controls; Tables 6.13–6.14.

Archive evidence: final_reviewer_closure/reference_tdse/tdse_three_level.csv and reference_grid_qcle/qcle_three_level.csv

4 — Numerical-noise and stochastic guards enforced

Status: Implemented and verified in the revised source

Correction: The absolute-plus-relative floor is 1e-12 + 1e-12 times the maximum level scale. Noise-limited, nonmonotone, and rapidly contracting nonasymptotic rows are rejected; MIDPOINT also requires resolution above independent-seed variation.

Final location: Thesis p. 80, Eq. (declared numerical-noise floor); Tables 6.7, 6.13, 6.14, and G.4.

Archive evidence: full-precision order-reason and verdict columns in the three reference/time-step CSVs

5 — Abstract claim calibrated

Status: Implemented and verified in the revised source

Correction: The abstract now says numerically controlled reference solutions and does not call them converged solutions.

Final location: Thesis p. i, Abstract.

Archive evidence: Thesis/Thesis.tex

6 — Independent clouds no longer called convergent

Status: Implemented and verified in the revised source

Correction: N=500, 1000, and 2000 are described as independently sampled, nonnested cloud enlargement; no deterministic cloud order is claimed.

Final location: Thesis p. 93, Section 6 independent-cloud enlargement; Table 6.8.

Archive evidence: final_reviewer_closure/support/independent_cloud_summary.csv

7 — Retrievable release record supplied

Status: Implemented and verified in the revised source

Correction: The final release identifies the archive filename, public release URL, source commit, archive SHA-256, checksum-index SHA-256, environment, manifests, and reproduction instructions.

Final location: Thesis p. 165, Appendix G, Section G.5 archive record.

Archive evidence: frozen_numerical_evidence_payload.zip; `https://github.com/sahandgit/gaussian_process_guided_mapping_quantumclassical_liouville_dynamics/releases/tag/thesis-final-2026-08-01`

8 — Visible typography repaired

Status: Implemented and verified in the revised source

Correction: The chapter range and radial-basis cross-validation phrase are kept unbroken; the dense Appendix G evidence remains complete.

Final location: Thesis Chapter 1 roadmap and Chapter 4 cross-validation text.

Archive evidence: Thesis/Thesis.tex

9 — Final response crosswalk added

Status: Implemented and verified in the revised source

Correction: This section maps each mandatory correction to its final thesis page/section/table and machine-readable archive source.

Final location: This response, Final mandatory corrections section.

Archive evidence: reviewer_data_audit/final_submission_document_manifest.json

Ten acceptance gates

Gate 1

Status: Addressed in the revised thesis

Response: Each chapter states its question, motivation, approach, and outcome in connected prose.

Revision in thesis: Chapter 1 opening argument (Section 1.1); opening and closing paragraphs of Chapters 2–7

Gate 2

Status: Addressed in the revised thesis

Response: The novelty statement is literature-bounded and application-specific.

Revision in thesis: Chapter 1, novelty paragraph in Section 1.1; Chapter 7, application-specific contribution section

Gate 3

Status: Addressed in the revised thesis

Response: Every retained scientific figure defines its quantity, averaging, uncertainty, and interpretation.

Revision in thesis: Chapter 6, Figures 6.1–6.5; figure-data crosswalk (final_reviewer_closure/figures/FIGURE_DATA_CROSSWALK.csv)

Gate 4

Status: Addressed in the revised thesis

Response: Density, gradient, and excess-action errors are reported on both the training cloud and an independent query cloud for every regularization and cloud size; the operator error is non-monotone with cloud size.

Revision in thesis: Chapter 6, Tables 6.2–6.4 and Figure 6.1; Appendix G, Tables G.1–G.3

Gate 5

Status: Addressed in the revised thesis

Response: Three endpoint values and two successive time-normalized differences are tested against the declared numerical-noise floor and independent-seed dispersion, so empirical order is withheld when either guard fails. TDSE and grid-QCLE use separate time and grid controls with explicit domains, discretizations, edge masses, and successive differences. The high-momentum TDSE grid case is identified as a boundary-adequacy control because all nominal levels use 8192 points.

Revision in thesis: Chapter 6, Tables 6.7 and 6.12–6.14; Appendix G, Table G.4; Appendix F reference settings

Gate 6

Status: Addressed in the revised thesis

Response: Clouds at N=500, 1000, and 2000 are independently sampled and non-nested, so the result is described as stochastic cloud-size sensitivity rather than deterministic convergence.

Revision in thesis: Chapter 6, independent-cloud enlargement section, Table 6.8

Gate 7

Status: Addressed in the revised thesis

Response: Paired MIDPOINT-minus-PBME physical-error differences do not consistently favour MIDPOINT, and the joint reliability requirements are not met.

Revision in thesis: Chapter 6, Table 6.15 and Figure 6.5; Appendix G, Tables G.7–G.8

Gate 8

Status: Addressed in the revised thesis

Response: The final code, evidence archive, checksum index, environment, manifests, and reproduction instructions are retrievable from
`https://github.com/sahandgit/gaussian_process_guided_mapping_quantumclassical_liouville_dynamics/releases/tag/thesis-final-2026-08-01`

Revision in thesis: Availability note of this response; Appendix G, Section G.5 archive record (release commit and payload SHA-256)

Gate 9

Status: Addressed in the revised thesis

Response: The title is synchronized as: Gaussian-Process Reconstruction of the Mapping-QCLE Excess Term: A Moving-Cloud Formulation and Failure Analysis.

Revision in thesis: Title page, hypersetup/PDF metadata, and this response document

Gate 10

Status: Addressed in the revised thesis

Response: Every I-, M-, and L-item is answered below with its revision location and scientific evidence.

Revision in thesis: The itemized I-, M-, and L-rows of this response, each with section, table/figure, page, and archive-source identifiers

I-items

I-1 — Complete three-seed manufactured result not shown

Status: Addressed in the revised thesis

Response: Emit every seed x N x on/off-support table with means and spreads. Density, gradient, and excess-action errors are reported on both the training cloud and an independent query cloud for every regularization and cloud size; the operator error is non-monotone with cloud size.

Revision in thesis: Chapter 6, manufactured-operator section

Evidence: Tables G.1–G.3 (complete per-seed values); aggregates in Tables 6.2–6.4 and Figure 6.1; thesis p. 146

I-2 — Time-step order conclusion cannot be reconstructed

Status: Addressed in the revised thesis

Response: One row per method/momentum/seed/observable/step triplet with both differences, guarded order, rejection reason, seed spread. Three endpoint values and two successive time-normalized differences are tested against the declared numerical-noise floor and independent-seed dispersion, so empirical order is withheld when either guard fails.

Revision in thesis: Chapter 6, time-step section

Evidence: Table G.4 (complete run-by-run evidence); summarized in Table 6.7; thesis p. 153

I-3 — Reference-convergence orders are unassigned

Status: Addressed in the revised thesis

Response: Recompute and classify all TDSE and grid-QCLE rows from separate three-level time and spatial/grid ladders under the declared guards. TDSE and grid-QCLE use separate time and grid controls with explicit domains, discretizations, edge masses, and successive differences. The high-momentum TDSE grid case is identified as a boundary-adequacy control because all nominal levels use 8192 points.

Revision in thesis: Chapter 6 and Appendix F

Evidence: Tables 6.13 and 6.14

I-4 — MIDPOINT central value at P0=20 absent

Status: Addressed in the revised thesis

Response: Report all four seed values, mean and SD for every estimator. Endpoint means, sample standard deviations, spreads, and Student-t intervals are reported in Table 6.9, and the four individual seed values for seeds 11, 29, 47, and 73 are shown in Figure 6.3 and the Delta-t=0.25 rows of Table G.4; MIDPOINT variability is far larger than PBME variability.

Revision in thesis: Chapter 6, replication section

Evidence: Table 6.9 (mean, SD, interval, spread); individual seed values in Figure 6.3 and the Delta-t=0.25 rows of Table G.4

I-5 — Identical-support pass incompletely reported

Status: Addressed in the revised thesis

Response: Report E1, E2, Einf at both momenta, or state which were not evaluated. The identical-cloud KDE/GP value-reconstruction criterion is met, while the thesis explicitly states that this does not validate derivatives or dynamics.

Revision in thesis: Chapter 6, common-support baseline

Evidence: Table 6.6

I-6 — 41 retained figures have no complete provenance

Status: Addressed in the revised thesis

Response: Retain only figures whose observable, normalization, independent realizations, uncertainty, and interpretation are completely defined. All retained figures define the observable, normalization, independent seed count, uncertainty bars, and sign convention in their captions.

Revision in thesis: Chapter 6, Figures fig:manufactured-operator-regularization through fig:paired-physical-reference-differences

Evidence: Figures 6.1–6.5

I-7 — Figure-dependent descriptions not interpretable

Status: Addressed in the revised thesis

Response: Attach a defined metric and numerical evidence to every retained comparison. The thesis uses five quantitative figures and fifteen reader-facing tables; every retained comparison has a defined metric, averaging rule, uncertainty description, and interpretation.

Revision in thesis: Chapter 6, Figures 6.1–6.5 and Tables 6.1–6.15

Evidence: Tables 6.1–6.15 and G.1–G.8, traced through final_reviewer_closure/table_data_crosswalk.csv

I-8 — Plotting thresholds undefined

Status: Addressed in the revised thesis

Response: State each threshold and its exact decision rule at the point of use. Every numerical threshold used for acceptance or display is stated with its rule. The numerical-noise, boundary-mass, negligible-tail-mass, and reconstruction thresholds are explicit in the relevant text, captions, and evidence CSVs.

Revision in thesis: Chapters 5–6 and Appendix F–G, relevant captions and table notes

Evidence: Tables 6.1–6.15 and G.1–G.8, traced through final_reviewer_closure/table_data_crosswalk.csv

I-9 — Diagnostics divide by undocumented |y0| tail

Status: Addressed in the revised thesis

Response: Report |y0| distribution and minimum, truncation rule, affected fraction of points and mass, threshold sensitivity. The signed-label threshold study reports excluded physical mass and signed/absolute effective sample sizes. No nontrivial negligible-mass plateau exists, so the tail is not assigned as a unique cause.

Revision in thesis: Chapters 5–7, tail-sensitivity analysis

Evidence: Tables 6.11 and G.5–G.6; thesis p. 158

I-10 — 'Applied source after stabilization' unspecified

Status: Addressed in the revised thesis

Response: Read the production branch and state the exact formula, or state unambiguously that applied source equals raw source. The applied excess source is distinguished from clipped or renormalized alternatives; the reported MIDPOINT results use the unprojected, nonconservative source under examination.

Revision in thesis: Chapter 6, applied-source controls

Evidence: Chapter 5 excess-source equations and Table 6.10

I-11 — 'Physically implausible' analytic observables not reported

Status: Addressed in the revised thesis

Response: Remove the unsupported analytic-observable claim rather than inventing uncomputed results. The unsupported analytic-GP physical-observable claim was removed; no uncomputed table is implied. Retained physical comparisons use the reported cloud estimators and controlled references.

Revision in thesis: Chapters 6–7, physical-comparison and claim-boundary sections

Evidence: Tables 6.1–6.15 and G.1–G.8, traced through final_reviewer_closure/table_data_crosswalk.csv

I-12 — 'Only weakly' undefined and conflicts with Table 6.6

Status: Addressed in the revised thesis

Response: Replace the global statement with per-estimator, per-momentum numbers. The independent sample size is four clouds, not the number of trajectories; intervals are described as small-sample sensitivity measures.

Revision in thesis: Chapter 6, replication section

Evidence: Table 6.9 (mean, SD, interval, spread); individual seed values in Figure 6.3 and the Delta-t=0.25 rows of Table G.4

I-13 — High-momentum agreement not quantitative

Status: Addressed in the revised thesis

Response: Seed-resolved L1/L2 field errors and observable errors, or take the escape. Paired MIDPOINT-minus-PBME physical-error differences do not consistently favour MIDPOINT, and the joint reliability requirements are not met.

Revision in thesis: Chapter 6, PBME versus MIDPOINT section

Evidence: Table 6.15; per-seed absolute errors in Tables G.7–G.8

I-14 — Exact reference settings outside the thesis

Status: Addressed in the revised thesis

Response: Reproduce grid, domain, dt, t_final, boundary rule in Appendix F. Reference domains, time steps, grids, boundary conventions, edge masses, and CFL controls are stated in Chapter 6 and Appendix F.

Revision in thesis: Chapter 6 and Appendix F

Evidence: Tables 6.12–6.14 and F.1

I-15 — Immutable configuration / versioned record not identified

Status: Addressed in the revised thesis

Response: Provide stable public access to the final research materials and state honestly whether a permanent archival identifier has been assigned. The final code, evidence archive, checksum index, environment, manifests, and reproduction instructions are retrievable from `https://github.com/sahandgit/gaussian_process_guided_mapping_quantumclassical_liouville_dynamics/releases/tag/thesis-final-2026-08-01`

Revision in thesis: Response availability note; Appendix G, Section G.5 archive and reproducibility record

Evidence: Appendix F and Tables F.1–F.2

I-16 — Thesis title not uniquely identified across artifact

Status: Addressed in the revised thesis

Response: One exact title in title page, PDF metadata, repository, response, paperwork. The title is synchronized as: Gaussian-Process Reconstruction of the Mapping-QCLE Excess Term: A Moving-Cloud Formulation and Failure Analysis.

Revision in thesis: Title page, PDF metadata, response, release

Evidence: Tables 6.1–6.15 and G.1–G.8, traced through final_reviewer_closure/table_data_crosswalk.csv

M-items

M-1 — Contribution not separated from known mathematical fact

Status: Addressed in the revised thesis

Response: State the application-specific chain, not the generic identifiability fact. The requested wording, evidence boundary, or numerical table is present.

Revision in thesis: Chapters 6–7 and Appendix F

Evidence: Tables 6.1–6.15 and G.1–G.8, traced through final_reviewer_closure/table_data_crosswalk.csv

M-2 — 'Projected density is represented' is misleading

Status: Addressed in the revised thesis

Response: Target is projected; the tested surrogate does not remain in its image. The four-field SEO image is derived and the measured leakage is large; projection is diagnosed but not enforced.

Revision in thesis: Chapters 4, 6, and 7

Evidence: Table 6.5

M-3 — 'Semi-discretization of the complete generator' too strong

Status: Addressed in the revised thesis

Response: Call it an attempted moving-cloud collocation of both formal generator terms. The requested wording, evidence boundary, or numerical table is present.

Revision in thesis: Chapters 6–7 and Appendix F

Evidence: Tables 6.1–6.15 and G.1–G.8, traced through final_reviewer_closure/table_data_crosswalk.csv

M-4 — 'Corrected method' implies correctness

Status: Addressed in the revised thesis

Response: Use 'MIDPOINT prototype' / 'tested excess-update branch'. Visible method-level correctness vocabulary was replaced by MIDPOINT prototype, source update, or tested excess-source branch.

Revision in thesis: Thesis source throughout

Evidence: Tables 6.1–6.15 and G.1–G.8, traced through final_reviewer_closure/table_data_crosswalk.csv

M-5 — 'Four checks form a controlled argument' overstates

Status: Addressed in the revised thesis

Response: Say the checks evidence failure under tested settings; list what stays unaudited. Failure is reported for the tested discretization; causal allocation remains explicitly nonunique.

Revision in thesis: Chapters 6–7

Evidence: Tables 6.1–6.15 and G.1–G.8, traced through final_reviewer_closure/table_data_crosswalk.csv

M-6 — 'Support refinement/convergence' used too freely

Status: Addressed in the revised thesis

Response: Call it an independent-cloud enlargement study; state convergence untested. Clouds at N=500, 1000, and 2000 are independently sampled and non-nested, so the result is described as stochastic cloud-size sensitivity rather than deterministic convergence.

Revision in thesis: Chapter 6, independent-cloud enlargement

Evidence: Table 6.8

M-7 — l2=0.01 control does not globally rule out regularization

Status: Addressed in the revised thesis

Response: State only: 0.05 -> 0.01 did not remove instability for P0=100, seed 11. All three regularizations are compared under paired manufactured clouds, with no claim that regularization alone cures the instability.

Revision in thesis: Chapter 6, regularization controls

Evidence: Tables 6.1–6.15 and G.1–G.8, traced through final_reviewer_closure/table_data_crosswalk.csv

M-8 — Formal second order is not demonstrated order

Status: Addressed in the revised thesis

Response: State assumptions unverified and positive production order not observed. Time-step changes are judged against seed dispersion and are not converted into formal convergence claims when unresolved.

Revision in thesis: Chapter 6, time-step section

Evidence: Table G.4 (complete run-by-run evidence); summarized in Table 6.7; thesis p. 153

M-9 — 'Qualitative visualization' does not admit untraceable images

Status: Addressed in the revised thesis

Response: Regenerate/trace, or remove images and all figure-dependent conclusions. No untraceable scientific figure remains.

Revision in thesis: Chapter 6, figure disposition

Evidence: Tables 6.1–6.15 and G.1–G.8, traced through final_reviewer_closure/table_data_crosswalk.csv

M-10 — Independent shape normalization conceals the failure

Status: Addressed in the revised thesis

Response: Label conclusions shape-only; show raw integral beside each. Raw normalization, trace, and energy are reported before display normalization; MIDPOINT exhibits severe drift whereas PBME remains stable on the reported scale.

Revision in thesis: Chapter 6, conservation section

Evidence: Table 6.10

M-11 — 'Less than ~1% per step' reads as a pointwise bound

Status: Addressed in the revised thesis

Response: State it is a ratio of global RMS norms and bounds nothing pointwise. The requested wording, evidence boundary, or numerical table is present.

Revision in thesis: Chapters 6–7 and Appendix F

Evidence: Tables 6.1–6.15 and G.1–G.8, traced through final_reviewer_closure/table_data_crosswalk.csv

M-12 — 'Not confined to negligible-label tails' unsupported

Status: Addressed in the revised thesis

Response: Supply quantile-resolved source contributions and tail-threshold sensitivity. The signed-label ratio, effective sample sizes, excluded mass, normalization, and energy are examined together across the threshold sweep.

Revision in thesis: Chapters 5–7

Evidence: Tables 6.11 and G.5–G.6; thesis p. 158

M-13 — Self-normalized GP normalization too prominent

Status: Addressed in the revised thesis

Response: Label as tautological; foreground the raw integral and GP-cloud discrepancy. Self-normalized unity is never used as conservation evidence.

Revision in thesis: Chapter 6, conservation section

Evidence: Table 6.10

M-14 — No explicit estimator hierarchy

Status: Addressed in the revised thesis

Response: Define each estimator once with validity domain and admissibility order. Chapter 6 states the hierarchy from raw invariants and reference errors to internal surrogate diagnostics before interpreting the data.

Revision in thesis: Opening of Chapter 6

Evidence: Tables 6.1–6.15 and G.1–G.8, traced through final_reviewer_closure/table_data_crosswalk.csv

M-15 — 'Anchor-cloud estimator is more defensible' too strong

Status: Addressed in the revised thesis

Response: Neither estimator is validated in the low-ESS regime. Low signed effective sample size is treated as loss of estimator reliability, not as validated physical information.

Revision in thesis: Chapters 6–7

Evidence: Tables 6.1–6.15 and G.1–G.8, traced through final_reviewer_closure/table_data_crosswalk.csv

M-16 — Undefined proximity language for the reference benchmark

Status: Addressed in the revised thesis

Response: Define by numerical field and observable errors with thresholds. The requested wording, evidence boundary, or numerical table is present.

Revision in thesis: Chapters 6–7 and Appendix F

Evidence: Tables 6.1–6.15 and G.1–G.8, traced through final_reviewer_closure/table_data_crosswalk.csv

M-17 — Causation attributed to the excess update, not the discretization

Status: Addressed in the revised thesis

Response: Attribute to the tested nonprojected, nonconservative MIDPOINT discretization. The negative conclusion is restricted to the tested product-GP moving-cloud MIDPOINT discretization, not the continuum QCLE excess term.

Revision in thesis: Chapters 6–7

Evidence: Tables 6.1–6.15 and G.1–G.8, traced through final_reviewer_closure/table_data_crosswalk.csv

M-18 — 'PBME reconstruction is faithful' broader than the gate

Status: Addressed in the revised thesis

Response: State exactly that the declared E1 gate passed on the tested support. The identical-cloud control isolates value reconstruction and is not generalized to derivative or propagation accuracy.

Revision in thesis: Chapter 6, common-support baseline

Evidence: Table 6.6

M-19 — Failure attribution not fully isolated

Status: Addressed in the revised thesis

Response: Describe as an evidence-supported pathway; causality not uniquely apportioned. The thesis presents a compatible failure pathway but explicitly avoids claiming a unique causal mechanism.

Revision in thesis: Chapters 6–7

Evidence: Tables 6.1–6.15 and G.1–G.8, traced through final_reviewer_closure/table_data_crosswalk.csv

M-20 — 'Reproducible basis' contradicts archive defects

Status: Addressed in the revised thesis

Response: Use 'documented but presently incomplete basis for future development'. Appendix G, Section G.5 gives a retrievable release record while the scientific chapters retain only the numerical methods and parameter scope needed to interpret the calculations.

Revision in thesis: Chapter 7 and Appendix F

Evidence: Tables 6.1–6.15 and G.1–G.8, traced through final_reviewer_closure/table_data_crosswalk.csv

M-21 — Objective broader than the tested construction

Status: Addressed in the revised thesis

Response: Name the single product-GP/moving-cloud/MIDPOINT construction and its 1D two-state test. The objective is restricted to one product-GP moving-cloud construction on a one-dimensional two-state benchmark.

Revision in thesis: Chapter 1 objective

Evidence: Tables 6.1–6.15 and G.1–G.8, traced through final_reviewer_closure/table_data_crosswalk.csv

M-22 — Chapter 6 opens with the wrong decision question

Status: Addressed in the revised thesis

Response: Begin with the four Chapter 1 acceptance criteria and the negative answer. Chapter 6 opens with the decisive physical reliability question and its controlled negative answer.

Revision in thesis: Opening of Chapter 6

Evidence: Tables 6.1–6.15 and G.1–G.8, traced through final_reviewer_closure/table_data_crosswalk.csv

M-23 — 'Full-density representation' ambiguous

Status: Addressed in the revised thesis

Response: Define at first use as a software architecture; rename to avoid implying exactness. Ambiguous full-density wording is absent.

Revision in thesis: Chapters 1, 4, 6, and 7

Evidence: Tables 6.1–6.15 and G.1–G.8, traced through final_reviewer_closure/table_data_crosswalk.csv

M-24 — 'Selected l2 = 0.01' needs qualification

Status: Addressed in the revised thesis

Response: Call it pilot-selected; repeat the operator test at production 0.05 and pilot 0.01. Complete manufactured density, gradient, and excess-action evidence is summarized over three independent cloud pairs for every tested regularization and cloud size.

Revision in thesis: Chapter 6, manufactured-operator section

Evidence: Tables G.1–G.3 (complete per-seed values)

M-25 — Completion counts confused with validation

Status: Addressed in the revised thesis

Response: Replace computational completion language with a numerical-design table organized by physical question and diagnostic. The numerical design table states the physical question, controlled variation, independent realizations, and diagnostic for every study.

Revision in thesis: Chapter 6, numerical design and hierarchy of evidence

Evidence: Table 6.1

L-items

L-1 — Use one method vocabulary

Status: Addressed in the revised thesis

Response: Terminology, definitions, claim calibration, local clarity, and the row-level audit trail were applied.

Revision in thesis: Thesis source and this reviewer response

Evidence: Tables 6.1–6.15 and G.1–G.8, traced through final_reviewer_closure/table_data_crosswalk.csv

L-2 — Define every estimator at first use and map it to one equation

Status: Addressed in the revised thesis

Response: Terminology, definitions, claim calibration, local clarity, and the row-level audit trail were applied.

Revision in thesis: Thesis source and this reviewer response

Evidence: Tables 6.1–6.15 and G.1–G.8, traced through final_reviewer_closure/table_data_crosswalk.csv

L-3 — Reserve evidentiary words for declared standards

Status: Addressed in the revised thesis

Response: Terminology, definitions, claim calibration, local clarity, and the row-level audit trail were applied.

Revision in thesis: Thesis source and this reviewer response

Evidence: Tables 6.1–6.15 and G.1–G.8, traced through final_reviewer_closure/table_data_crosswalk.csv

L-4 — Correct compound-word and hyphenation defects

Status: Addressed in the revised thesis

Response: Terminology, definitions, claim calibration, local clarity, and the row-level audit trail were applied.

Revision in thesis: Thesis source and this reviewer response

Evidence: Tables 6.1–6.15 and G.1–G.8, traced through final_reviewer_closure/table_data_crosswalk.csv

L-5 — Replace vague antecedents

Status: Addressed in the revised thesis

Response: Terminology, definitions, claim calibration, local clarity, and the row-level audit trail were applied.

Revision in thesis: Thesis source and this reviewer response

Evidence: Tables 6.1–6.15 and G.1–G.8, traced through final_reviewer_closure/table_data_crosswalk.csv

L-6 — Compress repetitive figure narration

Status: Addressed in the revised thesis

Response: Terminology, definitions, claim calibration, local clarity, and the row-level audit trail were applied.

Revision in thesis: Thesis source and this reviewer response

Evidence: Tables 6.1–6.15 and G.1–G.8, traced through final_reviewer_closure/table_data_crosswalk.csv

L-7 — Make the response document an audit trail

Status: Addressed in the revised thesis

Response: Terminology, definitions, claim calibration, local clarity, and the row-level audit trail were applied.

Revision in thesis: Thesis source and this reviewer response

Evidence: Tables 6.1–6.15 and G.1–G.8, traced through final_reviewer_closure/table_data_crosswalk.csv

Final scientific statement

The revision retains unfavourable and unstable outcomes rather than turning numerical completion into a success claim. Evidence is presented as physical observables, raw invariants, operator errors, uncertainty intervals, and controlled-reference comparisons. No internal surrogate diagnostic is presented as a physical-error estimate.

Availability note

The current research-code repository is `https://github.com/sahandgit/gaussian_process_guided_mapping_quantumclassical_liouville_dynamics`. The permanent identifier is `https://github.com/sahandgit/gaussian_process_guided_mapping_quantumclassical_liouville_dynamics/releases/tag/thesis-final-2026-08-01`.